Predictable Scale Dependence Makes Gradient Heavy Tails Removable: Scale-Normalized Sequential Learning

Anonymous Authors[1]

Abstract

We study online prediction on streams whose loss-gradient *scale* is strongly nonstationary, using a large financial dataset as a testbed. At a fixed predictor the *pooled* gradient-norm tail is genuinely heavy (Hill index $\hat{\alpha} \approx 2.4$), but *conditionally removable*: the *causally scale-normalized* gradient—what a learner acts on—is markedly lighter ($\hat{\alpha} \approx 2.9$–3.7). This is a property of *serial dependence*, not the marginal: a marginal-preserving order-shuffle destroys it, while the same normalization is tail-index preserving on iid streams of known index. The heavy marginal is volatility clustering—a bounded drift cannot move a tail index (Breiman, 1965), a predictable heavy-tailed scale can (Kesten/GARCH; Kesten 1973), and a GARCH surrogate with light innovations reproduces both the tail and its removal. In this regime plain OGD has no *robust* learning rate—bounded rates underfit while competitive rates diverge on scale spikes—and clipping to a *scale-dependent* threshold still inherits the drift. We propose *Scale-Normalized OMD* (SN-OMD): divide the gradient by a predictable robust scale and cap it at a *scale-free* constant M, so the per-step move is bounded by M and the iterates provably cannot blow up ($O(\sqrt{t})$ growth for any rate or realization; a stability, not convergence, guarantee). A dynamic-regret analysis attaining the per-regime rate $T^{1/p}$ ($p \in (1, 2]$) is deferred to the appendix, since the scale drift enters it as a genuine $\sqrt{T}$ factor. Empirically, a whole family of *bounded scale-free* updates stays bounded and accurate where OGD and scale-dependent clipping diverge; across ten disjoint windows with frozen hyperparameters every finite-cap member stays bounded ($0/10$ divergences) while the uncapped endpoint and OGD do not. No single member dominates across *both* protocols—SN-OMD, with a predictable block-median scale, is the most accurate per-row and normalized-GD per-timestep—so what singles out SN-OMD is a *predictable* scale, exactly what the dynamic-regret guarantee needs, not a uniform accuracy win.

1. Introduction

Financial markets are a canonical source of heavy-tailed, distribution-free data: returns and microstructure signals exhibit power-law tails, occasional data errors act as adversarial contamination, and assumptions such as sub-Gaussianity are untenable. Sequential prediction on such streams therefore calls for *distribution-free* guarantees that hold under weak moment assumptions only. A rich line of work in robust statistics shows that the empirical mean is a poor estimator under heavy tails, whereas estimators such as Catoni's M-estimator, median-of-means, and the trimmed mean achieve sub-Gaussian deviations assuming only a finite variance (Catoni, 2012; Devroye et al., 2016; Lugosi & Mendelson, 2019); in optimization, gradient clipping is the corresponding algorithmic primitive for heavy-tailed gradient noise (Zhang et al., 2020; Gorbunov et al., 2020; Nguyen et al., 2023).

This paper concerns a facet these lines mostly abstract away: on real streams the gradient *scale* is strongly nonstationary, and this—more than an intrinsically heavy tail—is what breaks existing methods. Measuring the gradient of an online linear predictor on the Jane Street market dataset, we find (i) the gradient *scale* drifts roughly $6\times$ across two months with overnight jumps; (ii) the *pooled* gradient-norm tail is genuinely heavy ($\hat{\alpha} \approx 2.4$) but *conditionally removable*—the *causally scale-normalized* gradient $\|g_t\|/s_t$, which is what a learner actually acts on, is much lighter ($\hat{\alpha} \approx 2.9$–3.7), and the lightening is a property of *serial dependence*, not of the marginal; and (iii) the *gradient* tail is heavier than either its feature or residual factor alone, an *interaction* effect from their co-movement. What the learner faces is thus the *predictable* part of a

[1]Anonymous Institution, Anonymous City, Anonymous Region, Anonymous Country. Correspondence to: Anonymous Author <anon.email@domain.com>.

Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.

nonstationary scale; the residual is dominated by causal-estimation cost, with no heavy tail separately resolvable above it (Table 1). The heavy pooled marginal is a *volatility-clustering* effect: the bounded $\approx 6\times$ drift cannot itself move a tail index (Breiman, 1965), but a predictable scale with a heavy-tailed conditional law can (the Kesten/GARCH mechanism, Kesten 1973)—the classical route to heavy tails in asset *returns* (ARCH/GARCH and the near-Gaussianity of volatility-standardized returns; Section 5). Our contribution is that it holds for the loss *gradients* of an online learner—as a feature$\times$residual interaction—and that it has an exploitable algorithmic consequence, developed below. This regime—a drifting scale against a moving comparator—is exactly where existing distribution-free online learning results, which are either in expectation, on bounded domains, or against a static comparator (Zhang et al., 2026; Zhang & Cutkosky, 2022), do not reach.

The consequences are sharp. Plain OGD has *no robust* learning rate on such a stream: rates small enough to stay bounded underfit, while any rate large enough to be competitive diverges the first time the gradient scale spikes in a turbulent regime (across ten windows its frozen rate diverges on $9/10$; Table 2). The textbook fix—clip the gradient at a threshold τ_t tracking the local scale—does not help under a drifting scale, because the clipped step is $\propto \min(\|g_t\|, \tau_t)$ and so still scales *with* the regime: a turbulent window produces large steps and diverges. We show this failure both empirically and as a structural property of any scale-dependent threshold.

Our approach. The fix is to decouple the step from the tail scale. *Scale-Normalized OMD* (SN-OMD) divides the gradient by a predictable robust scale s_t and caps the result at a *scale-free* constant M,

$$\hat{g}_t \;=\; \mathrm{clip}(g_t/s_t,\; M)\,, \qquad w_{t+1} \;=\; \Pi_{\mathcal{W}}\Big(w_t - \tfrac{\eta}{\sqrt{t}}\,\hat{g}_t\Big)\,.$$

The single factor $1/s_t$ replaces clipping's scale-*dependent* step bound Ms_t with a scale-*free* bound M. Because $\|\hat{g}_t\| \le M$ deterministically, the iterates grow at most as $O(\sqrt{t})$ (bounded on a bounded $\mathcal{W}$) for *any* learning rate or realization—an unconditional *stability* guarantee (not, on its own, convergence) that needs no moment assumption.

Contributions.

- **Heavy gradient tails are conditionally removable (Section 4).** On a large market dataset the pooled gradient tail is genuinely heavy ($\hat{\alpha} \approx 2.4$) but a learner with a *predictable* causal scale tracker faces a much lighter one ($\hat{\alpha} \approx 2.9$–3.7). Removability is a property of *serial dependence*: a marginal-preserving order-*shuffle* abolishes it, and the same normalization is tail-index preserving on iid streams of known index (so it is no artifact). The heavy marginal is volatility clustering (rank-based $Q_{100} \approx 7\times 10^5$)—a bounded drift cannot produce it (Breiman, 1965), a heavy-tailed predictable scale can (Kesten/GARCH), and a GARCH surrogate with light innovations reproduces both the tail and its removal. The effect recurs, in weaker form, on image-model training gradients (Section D.9). This is the paper's central finding.
- **Algorithm, stability, and a cap frontier (Sections 3 and 4.1).** We propose SN-OMD, prove its iterates cannot blow up ($O(\sqrt{t})$ growth on $\mathbb{R}^d$, bounded on a bounded domain; Theorem 3.1), and show it is a one-parameter family in the cap M interpolating normalized-GD ($M \to 0$) and uncapped scale-adaptive OGD ($M \to \infty$); a two-directional test predicts *when* the finite cap matters—non-divergence at large steps (against the uncapped endpoint), and an accuracy optimum only when the normalized tail is genuinely heavy.
- **A protocol-careful study, and deferred theory (Section 4).** Under causal standardization SN-OMD stays bounded and accurate (one of several co-leading bounded scale-free methods, not the single best; Table 4) where OGD and scale-dependent clipping diverge; it is invariant to per-row vs. per-timestep updates (no contemporaneous leakage) and its advantage vanishes on a genuinely stationary control. A per-regime dynamic-regret bound (Theorem C.2) is deferred to the appendix, honest about a $\sqrt{T}$ drift factor; a tested implementation is provided as anonymized supplementary material.

2. Problem Setup

Protocol. Learning proceeds in rounds $t = 1, \ldots, T$. At round t the learner plays $w_t \in \mathcal{W} \subseteq \mathbb{R}^d$ (a convex set of diameter D; the unbounded case is discussed in Section 3), observes a convex loss ℓ_t (for us the sample-weighted squared error $\ell_t(w) = \omega_t(\langle w, x_t\rangle - y_t)^2$), and receives a stochastic subgradient g_t with $\mathbb{E}[g_t \mid \mathcal{F}_{t-1}] = \bar{g}_t \in \partial\ell_t(w_t)$, where $(\mathcal{F}_t)$ is the natural filtration and $\varepsilon_t = g_t - \bar{g}_t$ is the noise.

Drifting heavy-tailed gradients. We assume only a *finite, time-varying* low-order moment of the subgradient itself—the quantity we measure and the algorithm acts on:

Assumption 2.1 (Drifting p-th moment)**.** For each t, $\mathbb{E}[\|g_t\|^{p_t} \mid \mathcal{F}_{t-1}] \le \sigma_t^{p_t}$ for some $p_t \in (1, 2]$, where the scale σ_t and the tail index p_t are *predictable* ($\mathcal{F}_{t-1}$-measurable) and may drift with t, and $p_t < 2$ (infinite variance) is permitted. Write $p := \inf_t p_t$. For the high-probability regret analysis (Section C) we further take the aggregates $S_1 := \sum_t \sigma_t$ and $\sup_t \sigma_t$ to admit *deterministic* almost-sure upper bounds, so that Freedman's a.s. variance condition (Lemma D.3) is met by a nonrandom bound. (This bounds $\bar{g}_t$ and the noise $\varepsilon_t = g_t - \bar{g}_t$ up to constants, and is exactly the tail we estimate in Section 4.)

No sub-Gaussianity, boundedness, or stationarity is assumed. *Distribution-free* here means the learner posits *no parametric or sub-Gaussian noise law*—only the finite, drifting p-th moment of Assumption 2.1. Two guarantees follow, at different knowledge costs. The unconditional *stability* guarantee (Theorem 3.1) uses *no* problem constant—not σ_t, p_t, D, or the comparator norm. The *regret* guarantee (Theorem C.2) is weaker in this respect: its tuned step is set from scale/path aggregates, of which p, the comparator norm, and the regime length are discharged by standard doubling/parameter-free wrappers (Section D), while the scale-path quantities W_s, P_T^s, S_1 enter as honest inputs rather than known constants. The regret analysis also requires a *predictable* scale tracker (Assumption C.1), which we do not treat as free but discharge under a mild drift model (Proposition D.5).

Dynamic regret. Performance is measured against a moving comparator $u_{1:T}$ with path length $P_T = \sum_t \|u_t - u_{t-1}\|$,

$$R_T(u_{1:T}) = \sum_{t=1}^{T} \big[\ell_t(w_t) - \ell_t(u_t)\big] \;\le\; \sum_{t=1}^{T} \langle \bar{g}_t, w_t - u_t \rangle, \tag{1}$$

the notion appropriate for regime-switching streams; static regret is the special case $u_t \equiv u$. We report the sample-weighted zero-mean $R^2 = 1 - \sum_i \omega_i (y_i - \hat{y}_i)^2 / \sum_i \omega_i y_i^2$ on data.

3. Scale-Normalized Online Mirror Descent

Algorithm. SN-OMD maintains a *predictable* robust estimate $s_t > 0$ of the gradient-norm scale (an $\mathcal{F}_{t-1}$-measurable function of past gradient norms) and takes a scale-invariant, constant-capped step (Algorithm 1). Writing $v_t = g_t / s_t$ and $\mathrm{clip}(v, M) = v \cdot \min\{1, M/\|v\|\}$, the update is $w_{t+1} = \Pi_{\mathcal{W}}(w_t - \frac{\eta}{\sqrt{t}}\,\mathrm{clip}(v_t, M))$. The scale s_t can be any predictable robust tracker; we use a winsorized exponential moving average of $\|g_t\|$ by default, and analyze a two-timescale "peak-hold" variant below.

Why normalize, not clip. Clipping to a scale-dependent threshold $\tau_t = c\, s_t$ gives a step $\propto \min\{\|g_t\|, c\, s_t\}$, which scales *with* the drifting s_t: a turbulent regime yields large steps and divergence. Dividing by s_t first gives a step $\propto \min\{\|g_t\|/s_t, M\}$, bounded by the *scale-free* constant M. The two differ by the factor $1/s_t$, and this single factor is what removes the scale-drift transmission.

Theorem 3.1 (Stability, unconditional). *For any inputs, $\|\hat{g}_t\| \le M$ deterministically. Hence on a bounded domain $\|w_t\| \le D$ for all t, and on $\mathbb{R}^d$ (step $\eta/\sqrt{t}$) $\|w_t\| \le \|w_1\| + 2M\eta\sqrt{t}$. Hence the iterates remain uniformly bounded (on $\mathcal{W}$) or grow at most as $O(\sqrt{t})$ (on $\mathbb{R}^d$) for any learning rate, scale drift, or realization—a* stability *guarantee, not by itself convergence, that uses no moment assumption.*

Algorithm 1 Scale-Normalized OMD (SN-OMD)

> **Input:** cap M, step η, predictable scale tracker $\mathcal{S}$; $w_1 = 0$.
> **for** $t = 1$ **to** T **do**
> play w_t; observe ℓ_t and stochastic subgradient g_t
> $s_t \leftarrow \mathcal{S}.\mathrm{scale}$ (predictable, uses g_t for $t' < t$)
> $\hat{g}_t \leftarrow \mathrm{clip}(g_t/s_t,\ M)$
> $w_{t+1} \leftarrow \Pi_{\mathcal{W}}\big(w_t - \frac{\eta}{\sqrt{t}}\hat{g}_t\big)$
> $\mathcal{S}.\mathrm{update}(\|g_t\|)$
> **end for**

This is the crucial contrast with scale-*dependent* clipping, whose per-step move $\propto s_t$ is bounded only by the drifting scale. The bound is a *family* property, not special to SN-OMD—any update whose step is capped by a scale-free constant satisfies it, including normalized-GD, a constant-threshold clip, and AdaGrad-Norm (Table 4). So Theorem 3.1 separates the bounded scale-free family from the scale-dependent methods; what singles out SN-OMD *within* that family is taken up in Section 4.

Dynamic regret (deferred to the appendix). Beyond stability, SN-OMD admits a high-probability *dynamic*-regret analysis (Section C): under a scale-tracking condition—a predictable s_t bracketing the local scale (Assumption C.1)—it attains the per-regime rate $T^{1/p}$ under only a finite p-th moment. We keep it in the appendix, not the headline, for two honest reasons. The bound is stated through the tracker's upward variation $W_s = s_1 + \sum_t (s_t - s_{t-1})_+$, which we *measure* to grow linearly with T on real data (Section 4.2); so the scale drift is a genuine $\sqrt{T}$ factor and the $T^{1/p}$ rate holds per regime, not globally. And the scale-tracking condition, though one-sided and empirically dischargeable (Proposition D.5), remains an assumption. What the body develops instead is the empirically-testable content—why normalizing beats clipping, and where the cap earns its keep.

4. Experiments

Setup and protocol. We use the Jane Street Real-Time Market Data Forecasting dataset (Jane Street, 2024): 47,127,338 rows over 1,699 trading days, 79 features, 39 symbols; target `responder_6`; metric the sample-weighted zero-mean R^2. The stability sweeps (Table 4, Figure 2) stream a continuous 150,000-row slice. We evaluate *progressive online* prediction—each round the learner plays w_t, then observes (x_t, y_t) and updates, the protocol our theory concerns—with features standardized *causally* (past-only; a full-sample standardizer inflates R^2 by look-ahead). We report two granularities, *per-row* and *per-timestep* (predict the (date, time) cross-section of ~ 9 symbols, then one aggregated update, carrying no contemporaneous cross-sectional information). **This is not the competition's protocol** (batched timesteps, day-lagged responders, held-out future window), so our R^2 is not comparable to leaderboard scores; we report it for cross-method comparison and the stability contrast. Tail measurements are learner-independent (gradients at $w^\star$): the pooled index from ~200,000 gradients, the daily-drift index from the full record.

Leakage controls for the R^2 magnitude. Because our prequential $R^2 \approx 0.28$ sits well above the competition's held-out leaderboard scores, we run a battery of look-ahead controls (Table 3 in Section A; all on the reported window). Permuting the (y_t, ω_t) pairs drives every method to $R^2 \le 0$ (SN-OMD -0.00: no label leakage); a static weighted ridge with *full* look-ahead reaches only 0.02 and a per-day look-ahead oracle 0.18, both below online SN-OMD's 0.28, so the accuracy is carried by fast within-day adaptation that no fixed-partition linear oracle matches; and the weights learned here, applied unchanged to a disjoint later window, give $R^2 = -1.7$ (that window's own prequential is $+0.22$). The reported R^2 is thus a *tracking* score on a fast-drifting stream, not a transferable static fit—itself evidence for the nonstationarity we study, and why it is not comparable to the competition's protocol.

Table 1. **The lightening is order-one and estimator-stable; the residual is causal-estimation cost.** Hill $\hat{\alpha}$ of the normalized $\|g_t\|$ at three tail fractions k (larger = lighter). The deployed causal tracker lightens the raw pooled tail by $+1.0$ to $+1.5$; the *same* estimator run non-causally recovers only $\sim +0.3$ more ($+0.14$–$+0.38$ over k and estimator). The GARCH surrogate has Gaussian innovations (*zero* residual tail) and is calibrated to the real raw index; after the same causal tracking it reads $\hat{\alpha} \approx 4.1$–4.8 across the three thresholds shown (widening to ≈ 3.9 at lower GARCH persistence; Figure 1) against a true-σ_t oracle of $\approx$ 8–10, so causal estimation *alone* costs several index points. The real 3.73 sits *just below* this zero-residual band at every setting we ran: the normalized real tail is at most marginally *heavier* than a null built to have none, and the gap ($\lesssim 0.8$) is no larger than the surrogate's own sensitivity to its persistence. We therefore cannot separate a genuine residual conditional tail from a scale process slightly less predictable than GARCH(1,1)—but either reading gives $\hat{\alpha} \approx 3.7$ ($p > 2$), so the residual is dominated by causal-estimation cost and the $(1, 2]$ apparatus does not bind here.

Normalization of $\|g_t\|$	k=.005	k=.01	k=.02
Real gradient stream:			
raw pooled (none)	2.55	2.43	2.32
causal winsorized-EMA *(deployed)*	4.02	3.73	3.36
centered EMA (non-causal twin)	4.16	4.05	3.58
trailing median (causal)	3.24	2.87	2.60
centered median (non-causal)	3.62	3.18	2.81
GARCH surrogate (Gaussian innov.; zero residual tail):			
raw pooled (none)	2.54	2.48	2.44
causal winsorized-EMA	4.81	4.48	4.08
true-σ_t oracle	10.3	9.05	7.75

The pooled tail is real, but conditionally removable. At the fixed optimum $w^\star$, the *pooled* gradient-norm tail index is $\hat{\alpha} \approx 2.43$, and it is an *interaction* effect: the residual and feature norms each have $\hat{\alpha} \approx 4$, but their product—the gradient—has a heavier tail because extreme features and residuals co-occur (Figure 5a). This heaviness is genuine, but *conditionally removable*: dividing by a causally-tracked robust scale s_t—the quantity SN-OMD actually acts on—lightens the tail markedly, to $\hat{\alpha} \approx 3.7$ (causal winsorized-EMA s_t), ≈ 3.2 (non-causal median), or ≈ 2.9 (trailing median), i.e. toward finite higher moments. The lightening is positive at every threshold and estimator (Table 1), and what remains is dominated by *causal-estimation cost* rather than a resolvable residual tail: isolating causality (the *same* estimator run non-causally) recovers only $\sim +0.3$, and on a GARCH surrogate with Gaussian innovations—*zero* residual tail by construction—the same causal tracker still reads $\hat{\alpha} \approx 3.9$–4.5 (depending on persistence) against a true-σ_t oracle of ≈ 9 (Table 1). The real 3.73 sits *below* this zero-residual band, so the normalized real tail is, if anything, marginally *heavier* than a null with nothing left to resolve; but the shortfall is of the same order as the surrogate's sensitivity to its own persistence, so we cannot separate a genuine residual conditional tail from a scale merely less predictable than GARCH(1,1). Either way the residual is dominated by causal-estimation cost (the surrogate's own oracle-vs-causal gap is ≈ 4.5 index points), and $\hat{\alpha} \approx 3.7$ leaves $p > 2$. What makes it removable is *serial dependence*: the gradient magnitude is strongly clustered—rank (Spearman) autocorrelation ≈ 0.16 at lag 100, rank-based Ljung–Box $Q_{100} \approx 7 \times 10^5$ (ranks are bounded, so the χ^2 is valid; a shuffled control gives ≈ 116), of which the $\approx 6\times$ daily drift (Figure 5b) is only the low-frequency component. Three controls (Figure 1; `research_null_normalization.py` in the anonymized supplement) pin this down: (i) a marginal-preserving order-*shuffle* leaves the pooled index unchanged but destroys the lightening, so the tracker exploits temporal *order*, not the marginal; (ii) the same causal normalization is tail-index *preserving* on iid streams of known index $(2.0, 2.4, 3.0)$, ruling out an estimator artifact; and (iii) a GARCH surrogate with *light* (Gaussian) innovations reproduces both the pooled $\hat{\alpha} \approx 2.4$ and its removal. The heavy *marginal* is *not* manufactured by the bounded $\approx 6\times$ drift—a bounded scale factor cannot move a tail index (Breiman, 1965)—but by a predictable scale whose conditional law is heavy-tailed (the Kesten/GARCH mechanism, Kesten 1973; a related route is the subordination of Clark 1973, Section 5). So the tail a learner faces is set by the *predictable* part of the scale. (Per-day $\hat{\alpha}$ dips below 2 on a minority of days, but single-day Hill estimates are noisy and we do not rely on them.) We note that the per-regime budget of Section B is dominated by exactly these pooled-$\alpha < 2$ days; since those regimes are themselves a pooled-tail artifact—they largely disappear once the gradient is normalized (the tail the learner sees has $p \approx 2$)—we treat that budget only as qualitative motivation for why a moving comparator under drift is hard, not as a measured quantity.

Scale-invariant updates dominate scale-dependent ones. Streaming one learner through the multi-regime window and sweeping the learning rate (Figure 2, Table 4), the methods split cleanly by whether their step responds to the gradient *scale*. *Scale-dependent* methods—plain OGD and the robust clippers (AdaptiveClip, RobustOMD), whose threshold $\tau_t = c\, s_t$ rescales with the regime—are fragile: they stay bounded only to $\mathrm{lr} \approx 10^{-2}$ and reach best $R^2 \approx 0.01$–0.14, because any larger rate diverges when the scale spikes (Figure 3). The *scale-invariant* family—normalized-GD, SN-OMD, uncapped scale-adaptive OGD, a constant-threshold fixed-τ clip, and AdaGrad-Norm—stays bounded far higher and reaches $R^2 \approx 0.08$–0.38 across the two protocols. Per-timestep batching lifts OGD from 0.02 to 0.14 (averaging the ~ 9 correlated symbols per timestep) but leaves it fragile; SN-OMD is essentially unchanged ($0.285 \to 0.309$), so its accuracy is not a contemporaneous-information artifact, whereas uncapped scale-adaptive OGD *halves* ($0.162 \to 0.080$: aggregating enlarges the untruncated step a finite cap would contain, forcing a lower stable rate—its frozen-rate consequence returns with the replication).

What the tracked scale and the cap buy—and what they do not. A whole *family* of bounded scale-free updates—SN-OMD, normalized-GD, a constant-threshold fixed-τ clip, and AdaGrad-Norm—closes the accuracy gap to the divergent baselines, and *no single member dominates*. On the reported window the two top entries use *neither* a tracked scale nor a cap (fixed-τ clip per-step, AdaGrad-Norm per-row), but neither ranking survives replication (Table 2 below): the winner differs by protocol and no member leads on both, so what earns the gain is *normalization in general*—any bounded scale-free step—of which SN-OMD is one instance. The family is not homogeneous, though: stable learning-rate ranges span two orders of magnitude (fixed-τ to $\mathrm{lr} \approx 0.2$, SN-OMD and normalized-GD to 2, AdaGrad-Norm to 10), so "bounded scale-free" is necessary for stability while the *ceiling* depends on how the step behaves *within* the bound.

What singles out SN-OMD is therefore narrower than accuracy, and we state it as such. *(i) Predictability:* its scale s_t is a *predictable* tracker, which is what makes the dynamic-regret bound (Theorem C.2) go through—normalized-GD's instantaneous $1/\|g_t\|$ is not predictable and discards magnitude every round; the payoff is the (per-regime) bound, not a measured accuracy edge. *(ii) The cap:* a finite M bounds the step directly, but non-divergence is *not* unique to it— AdaGrad-Norm stays bounded to a *wider* $\mathrm{lr} \leq 10$ with no cap. What the cap distinctly buys is an interior accuracy optimum when the *normalized* tail is genuinely heavy (Section 4.1); on Jane that tail is indistinguishable from causal-estimation cost, so the optimum is flat and the cap-family members are close. SN-OMD's block-bootstrap standard errors ($\pm.079/.056$) run $\approx$3$\times$ normalized-GD's ($\pm.024/.021$), which the ten-window replication traces to the winsorized-EMA tracker rather than the cap: the block-median scale removes it (per-row std $0.11 \to 0.04$; Table 2 and Section 4.2).

Replication across ten disjoint windows. A single contiguous window cannot settle a claim that is *about* regime dependence, so we re-run the entire comparison on *ten* disjoint 150,000-row windows spanning the full 1,699-day record, with every method's hyperparameters *frozen* at its window-1 tuning—the realistic tune-once-then-deploy setting (Table 2). Two findings are sharp. *(i) The stability partition replicates and hardens:* every finite-cap bounded scale-free method stays bounded on all ten windows ($0/10$ divergences), while the uncapped $M \to \infty$ endpoint diverges on $6/10$ (per-row) and plain OGD's frozen rate diverges on $9/10$ (per-step). The uncapped endpoint's $0/10$ per-step is not batching conferring robustness but a lower frozen rate: its per-timestep window-1 optimum sits at $\mathrm{lr}{=}0.2$ (vs. 0.5 per-row), so it underfits ($0.09 \pm .02$) instead of diverging, while the aggressive per-row rate blows up—the finite cap, by contrast, is stable at every rate on *both* protocols. So "OGD has no robust rate" and "the cap buys unconditional stability" hold across the whole record, not merely on the reported window. *(ii) No bounded member dominates, and which one is best is tracker-dependent:* per-row the winner rotates window to window (fixed-τ, AdaGrad-Norm, and normalized-GD each lead some), and per-step normalized-GD is the most robust (0.29 ± 0.02, best on $8/10$). With its *deployed* winsorized-EMA tracker SN-OMD stays bounded but is the most *variable* member (per-row 0.14 ± 0.11, negative on one window); swapping in the block-median tracker (Section 4.2) removes that variance and makes SN-OMD the *best* bounded method per-row (0.31 ± 0.04, never negative, $0/10$ divergences). So the excess variance is the EMA tracker's, not the cap's—but the robust-and-accurate tracker is the one *without* a proven W_s bound (Section 4.2), the same guarantee-versus-accuracy gap stated there. What generalizes unconditionally is *bounded scale-free stability*; *which* bounded member is most accurate depends on the tracker and protocol—now shown on the whole record, not a slice.

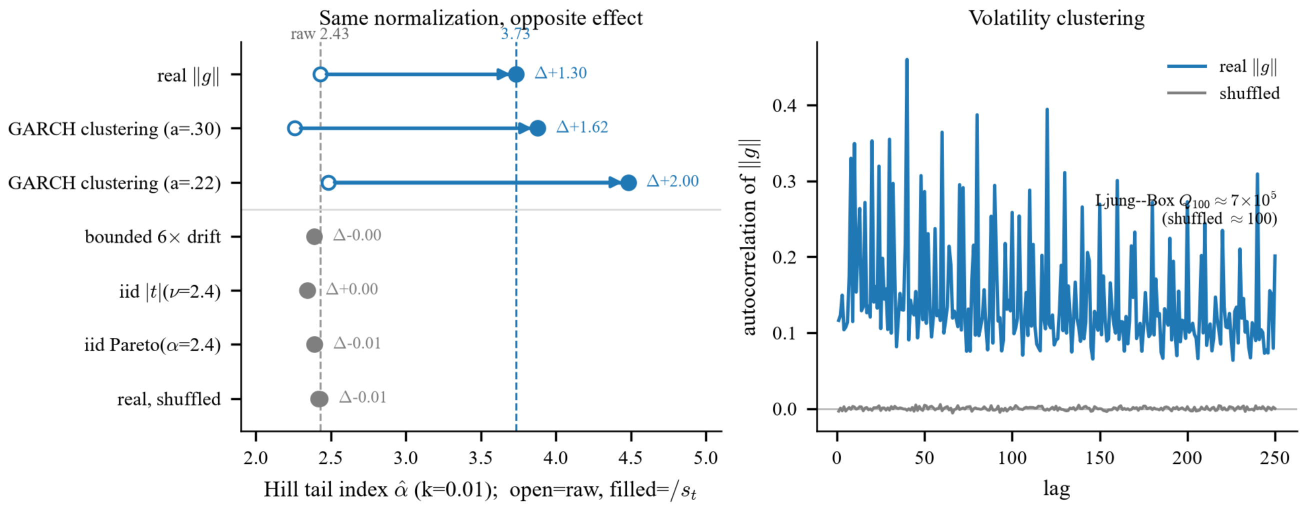

Figure 1. **The lightening is removability, not manufacture.** (a) Hill index before (open) and after (filled) causal scale-normalization, k=0.01. The *same* operation lightens the real gradient tail (2.43→3.73) and a light-innovation GARCH surrogate ($\approx$ 2.3→3.9–4.5), but leaves an order-*shuffle*, iid heavy streams of known index, and a bounded 6× drift essentially unchanged ($\Delta\approx 0$)—what is removed is serial dependence, not the marginal, and it is no estimator artifact. (b) Rank autocorrelation of $\|g_t\|$ decays slowly (the volatility clustering the tracker exploits) and vanishes under shuffling.

Table 2. **Ten-window replication, hyperparameters frozen from window 1** (Section 4; ten disjoint 150k-row windows across the 1,699-day record). Mean±std weighted R^2 and divergence count (a method "diverges" on a window if its peak rolling loss blows up at the frozen rate). Every finite-cap bounded scale-free method stays bounded on all ten windows; the uncapped endpoint and plain OGD do not. *No single member dominates across both protocols*: SN-OMD with the block-median tracker leads per-row (0.31±.04) and normalized-GD per-timestep (0.29 ± .02). SN-OMD is the only method shown with two tracker variants; its block median was selected on window 1 as the most accurate tracker, so this per-row lead is a best-case tracker choice, whereas every baseline uses a single fixed configuration.

	per-row		per-step	
Method	mean±std	div	mean±std	div
OGD	$0.00 \pm .04$	0/10	—	9/10
Scale-adaptive OGD ($M\to\infty$)	—	6/10	$0.09 \pm .02$	0/10
Normalized-GD ($M\to 0$)	$0.18 \pm .01$	0/10	$\mathbf{0.29} \pm .02$	0/10
Fixed-τ clip	$\mathbf{0.21} \pm .03$	0/10	$0.20 \pm .13$	0/10
AdaGrad-Norm	$0.20 \pm .07$	0/10	$0.19 \pm .03$	0/10
SN-OMD (M**=5, ours)**	$0.14 \pm .11$	0/10	$0.13 \pm .14$	0/10
+ *block-median tracker*	$\mathbf{0.31} \pm .04$	0/10	$0.20 \pm .13$	0/10

Negative control. If the advantage is really about scale drift, it should vanish when the scale is constant. Imposing genuine stationarity in the synthetic harness ($\sigma_t \equiv 1$, p=2, static reachable comparator—the one stream where stationarity can be enforced), tuned SN-OMD is if anything *worse* than tuned OGD (steady excess loss 6.6 vs. 5.6×10^{-4} over 400 seeds, an 18% gap the wrong way): the advantage disappears as the thesis predicts (a 100k-row Jane slice is *not* a valid stationarity control—it still has $\sim$ 30% within-day drift—so its lead there is expected).

4.1. The cap M interpolates normalized-GD and scale-adaptive OGD

SN-OMD is a one-parameter family in the cap M: when $\|g_t\|/s_t > M$ the scale cancels ($\text{clip}(g_t/s_t, M) = M\, g_t/\|g_t\|$), so $M \to 0$ is *normalized-GD* and $M \to \infty$ is uncapped *scale-adaptive OGD*. Every *finite-*M member satisfies Theorem 3.1 ($\|\hat{g}_t\| \le M$); the uncapped endpoint is the lone exception and is exactly the member that diverges. What a finite M distinctly buys over the normalized-GD endpoint is that it *preserves gradient magnitude within the bulk* while truncating only the tail, whereas normalized-GD discards magnitude every round. Sweeping M turns the cap into a *mechanism test* (Section D.8): on a synthetic stream with an *imposed* heavy normalized tail an interior optimum appears (the best cap beats both endpoints), while on Jane—where the normalized tail is indistinguishable from causal-estimation cost (Table 1)—there is *no* meaningful interior optimum and the three members are comparable. So the cap buys *accuracy* only when the tail survives normalization, and *stability unconditionally*. On synthetic streams with a *known* Student-t index, plain OGD's cumulative-regret exponent climbs past 1 (divergent) as the tail heavies while SN-OMD stays sublinear (Section D.8), so the tail is the mechanism, not a learning-rate artifact.

4.2. Discharging the scale-tracking assumption

The regret bound (Theorem C.2) rests on Assumption C.1: a *predictable* scale s_t brackets the local noise scale. We discharge it empirically on the real gradient-scale process (200,000 norms at $w^\star$; Figure 4, Table 5, and Section D.10). The requirement is one-sided *in what matters* (Section 3

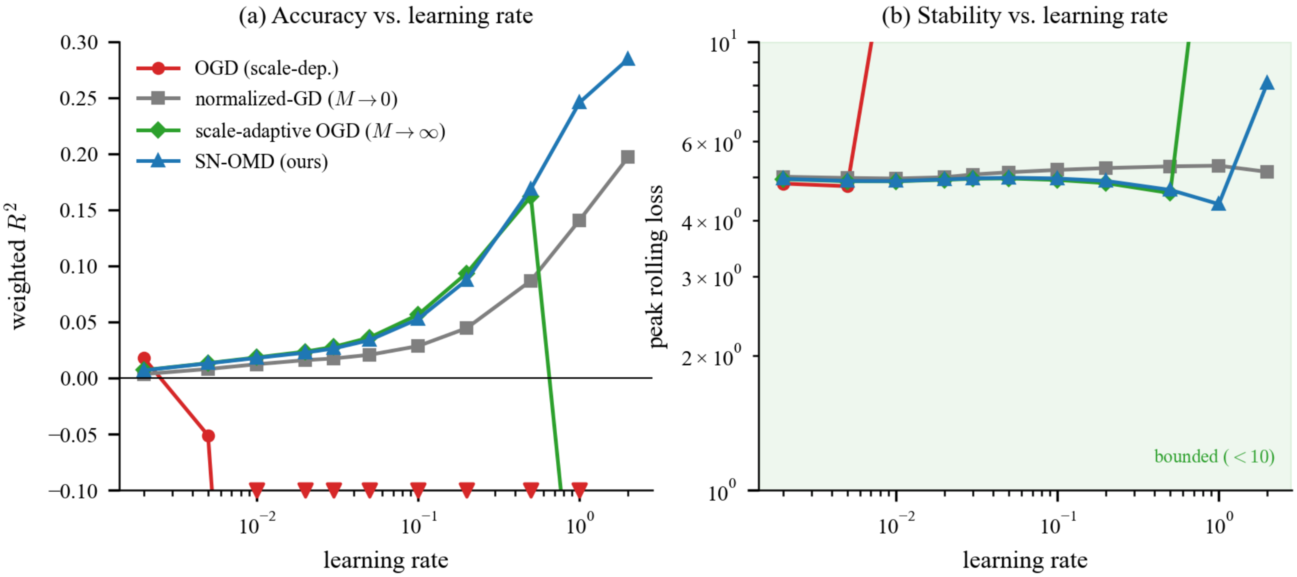

Figure 2. **Main result** (continuous multi-regime stream, *causal* standardization, per-row). The cap family: OGD (scale-dependent) vs. the $M\to 0$ endpoint (normalized-GD), the $M\to\infty$ endpoint (uncapped scale-adaptive OGD), and SN-OMD (M=5). (a) Weighted R^2 vs. learning rate: OGD cliffs above lr $= 10^{-2}$ (off-scale), the scale-invariant methods climb to $R^2 \approx 0.2$–0.3 (SN-OMD and normalized-GD comparable, intervals overlapping; Table 4), and uncapped scale-adaptive OGD diverges at large rates. (b) Peak rolling loss: OGD and uncapped scale-adaptive OGD explode (10^2–10^{30}), while normalized-GD (≈ 5) and SN-OMD (single-digit, ≈ 8 at its top stable lr=2) stay bounded.

and Assumption C.1)—hold the lower bracket $s_t \geq \frac{1}{2}\sigma_t$ everywhere (the stability-critical half) while keeping the upward variation W_s (entering Theorem C.2 as $\sqrt{W_s}$) near the true drift V_σ^+; the upper bracket enters only the regret *constant*, never divergence. A single-timescale trailing median cannot do both (a fast window churns on spikes, a slow one lags regime onsets), whereas a two-timescale "peak-hold" envelope—react up on a short robust median, release down geometrically—holds the lower bracket at 100% of rounds at the *smallest* upward variation, $W_s \approx 6.6\, V_\sigma^+$, at the benign cost of a 24% over-estimate. So the stability-critical half of Assumption C.1 is discharged.

The drift is a genuine rate; the tracker controls only its constant. W_s grows *linearly* in T (Table 6), so $\sqrt{W_s} \sim \sqrt{T}$ is genuine and Theorem C.2's $T^{1/p}$ is *per-regime* (globally the drift adds a $\sqrt{T}$ factor). A regime-timescale tracker shrinks the *constant* by $\sim 250\times$ at our horizon but not the exponent (Remark C.4); escaping the $\sqrt{T}$ needs a switching bound in the regime count, left open (Section D.7). The most accurate per-row tracker on Jane is a block median; stress-tested against the circular block bootstrap its edge over normalized-GD is real yet narrow and protocol-specific (Section D.10), so the "ties normalized-GD" headline stands where measured most conservatively. Past the tuned range both trackers degrade without diverging ($O(\sqrt{t})$ bound; Section D.10).

5. Related Work

Scale mixtures and volatility models. That a drifting scale makes conditionally light-tailed data look unconditionally heavy-tailed is classical *for asset returns*: Clark's subordinated-process model (Clark, 1973), the ARCH/GARCH family (Engle, 1982; Bollerslev, 1986), and the finding that returns *standardized by realized volatility* are approximately Gaussian (Andersen et al., 2001). Our pooled-vs-normalized separation is the same phenomenon; we claim no novelty for the mechanism. What is new is (i) that it holds for the *loss gradients* of an online learner—so it governs *optimization*, not just modeling; (ii) that the gradient's heavy tail is an *interaction* effect (feature × residual co-movement), heavier than either factor alone (Figure 5); and (iii) the algorithm-design consequence—the scale-normalized update and its cap-interpolation frontier (Section 4.1)—which the volatility literature does not address.

Robust estimation and clipping. Robust estimators (Catoni, median-of-means, trimmed mean) achieve sub-Gaussian deviations under a finite variance (Catoni, 2012; Devroye et al., 2016; Lugosi & Mendelson, 2019), imported into sequential decision-making by Bubeck et al. (2013); in optimization, gradient clipping yields high-probability convergence under heavy-tailed noise (Zhang et al., 2020; Gorbunov et al., 2020; Nguyen et al., 2023). Closest to ours, Cutkosky & Mehta (2021) *normalize and clip* for high-probability non-convex convergence under $\mathbb{E}\|g_t\|^p < \infty$; we differ in normalizing by a *predictable* tracked scale (not the current gradient) and in targeting *dynamic* regret under a *drifting* scale rather than static convergence. Our normalization is thus a streaming, scale-decoupled variant whose value is specifically stability under that drift.

Heavy-tailed online learning. Zhang et al. (2026) show plain OGD is optimal *in expectation* on a bounded domain, and Zhang & Cutkosky (2022) give parameter-free *high-probability* static regret $\tilde{O}(\|u\|T^{1/p})$ over unbounded domains, whose central obstacle—exponentially large iterates—our cap sidesteps. Both are static; the dynamic, nonstationary regime we target sits outside them, and our budget heuristic (Section B) suggests why it is harder (a proper P_T-constrained lower bound remains open).

Dynamic, scale-free, and adaptive OL. Dynamic regret against a moving comparator goes back to Zinkevich (2003); normalized and scale-free methods adapt to an unknown gradient scale (Levy, 2017; Orabona & Pál, 2018); strongly-adaptive and parameter-free wrappers adapt to unknown interval/regime structure (Daniely et al., 2015; Cutkosky, 2020), which we use as a black box to remove knowledge of the regime length.

Adaptive optimizers. Coordinate-wise adaptive methods (AdaGrad, RMSProp, Adam) also rescale the gradient by a running second moment, so a natural question is whether they already handle a drifting scale. The answer splits on whether the preconditioner is *monotone*. The *EMA-denominator* members (RMSProp, Adam) divide by $\sqrt{\mathrm{EMA}(g_t^2)}$, which *itself* spikes with a heavy-tailed gradient (numerator and denominator share the round); with no scale-*free* cap a single extreme gradient still yields an unbounded step, so they exhibit the divergence documented in Section 4. *AdaGrad-Norm* instead divides by the *monotone* accumulator $\sqrt{\sum_s \|g_s\|^2}$, which bounds its step by η; head-to-head on Jane (Table 4) it stays bounded and is accuracy-competitive (best per-row), placing it with the bounded scale-free family. What it lacks is a *predictable* scale and the dynamic-regret guarantee (Theorem C.2); our unconditional stability (Theorem 3.1) comes from capping the *normalized* gradient at M.

6. Conclusion

On real streams the difficulty is dominated by a strongly *nonstationary* gradient scale, whose *predictable* part is what the learner faces: the pooled gradient looks heavy-tailed (volatility clustering of a heavy-tailed conditional scale, not something the bounded drift can produce) while the causally normalized gradient is much lighter. SN-OMD normalizes and caps at a scale-free constant, so its iterates provably cannot diverge ($O(\sqrt{t})$ for any rate) and attain $T^{1/p}$ *per regime* (the drift is a genuine $\sqrt{W_s} \sim \sqrt{T}$ factor). Across ten disjoint windows the *bounded scale-free* family stays bounded where OGD and scale-dependent clipping diverge, with no member dominating across protocols (Table 2); what distinguishes SN-OMD is a *predictable* tracker—which the dynamic-regret guarantee requires—not an accuracy win.

Limitations. SN-OMD is not fully tuning-free (setting-dependent learning rate; untuned M, decay, winsorization; though its stable range far exceeds the baselines'). The $W_s = O(V_\sigma^+)$ guarantee covers only the two-timescale envelope (Assumption C.1), not the deployed EMA or block-median trackers (supported only empirically, Section 4.2); RMSProp/Adam are argued to diverge but not run head-to-head (AdaGrad-Norm and a constant clip are); and validation beyond our two domains is warranted. Finally, on this stream the normalized residual tail is dominated by *causal-estimation cost*: a zero-residual GARCH surrogate reads $\hat{\alpha} \approx 3.9$–4.8 after the same causal tracking and the real 3.73 sits just below it (Table 1), so we cannot separate a genuine residual tail from a scale less predictable than GARCH(1,1)—either way $\hat{\alpha} \approx 3.7$ leaves $p > 2$, so both the $p \in (1, 2]$ machinery and the cap's tail-truncation are worst-case guarantees with no purchase here (the cap's role is the stability of Theorem 3.1).

Impact Statement

This paper presents work whose goal is to advance the field of machine learning. There are many potential societal consequences of our work, none of which we feel must be specifically highlighted here.

References

Andersen, T. G., Bollerslev, T., Diebold, F. X., and Ebens, H. The distribution of realized stock return volatility. *Journal of Financial Economics*, 61(1):43–76, 2001.

Bollerslev, T. Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3):307–327, 1986.

Breiman, L. On some limit theorems similar to the arc-sin law. *Theory of Probability & Its Applications*, 10(2): 323–331, 1965.

Bubeck, S., Cesa-Bianchi, N., and Lugosi, G. Bandits with heavy tail. *IEEE Transactions on Information Theory*, 59 (11):7711–7717, 2013.

Catoni, O. Challenging the empirical mean and empirical variance: A deviation study. *Annales de l'Institut Henri Poincaré, Probabilités et Statistiques*, 48(4):1148–1185, 2012.

Clark, P. K. A subordinated stochastic process model with finite variance for speculative prices. *Econometrica*, 41 (1):135–155, 1973.

Cutkosky, A. Parameter-free, dynamic, and strongly-adaptive online learning. In *International Conference on Machine Learning (ICML)*, pp. 2250–2259, 2020.

Cutkosky, A. and Mehta, H. High-probability bounds for non-convex stochastic optimization with heavy tails. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 34, pp. 4883–4895, 2021.

Daniely, A., Gonen, A., and Shalev-Shwartz, S. Strongly adaptive online learning. In *International Conference on Machine Learning (ICML)*, pp. 1405–1411, 2015.

Devroye, L., Lerasle, M., Lugosi, G., and Oliveira, R. I. Sub-Gaussian mean estimators. *The Annals of Statistics*, 44(6):2695–2725, 2016.

Engle, R. F. Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation. *Econometrica*, 50(4):987–1007, 1982.

Gorbunov, E., Danilova, M., and Gasnikov, A. Stochastic optimization with heavy-tailed noise via accelerated gradient clipping. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.

Jane Street. Jane Street real-time market data forecasting. Kaggle competition, 2024. https://www.kaggle.com/competitions/jane-street-real-time-market-data-forecasting.

Kesten, H. Random difference equations and renewal theory for products of random matrices. *Acta Mathematica*, 131: 207–248, 1973.

Levy, K. Y. Online to offline conversions, universality and adaptive minibatch sizes. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

Lugosi, G. and Mendelson, S. Mean estimation and regression under heavy-tailed distributions: A survey. *Foundations of Computational Mathematics*, 19(5):1145–1190, 2019.

Nguyen, T. D., Ene, A., and Nguyen, H. L. High probability convergence of clipped-SGD under heavy-tailed noise. *arXiv preprint arXiv:2302.05437*, 2023.

Orabona, F. and Pál, D. Scale-free online learning. *Theoretical Computer Science*, 716:50–69, 2018.

Zhang, J. and Cutkosky, A. Parameter-free regret in high probability with heavy tails. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022. arXiv:2210.14355.

Zhang, J., He, T., Sra, S., and Jadbabaie, A. Why gradient clipping accelerates training: A theoretical justification for adaptivity. In *International Conference on Learning Representations (ICLR)*, 2020.

Zhang, Z., Zhang, M., and Zhou, Z. Online convex optimization with heavy tails: Old algorithms, new regrets, and applications. In *Algorithmic Learning Theory (ALT)*, 2026. arXiv:2508.07473.

Zinkevich, M. Online convex programming and generalized infinitesimal gradient ascent. In *International Conference on Machine Learning (ICML)*, pp. 928–936, 2003.

Software and Data

The accompanying anonymized code provides the `ScaleNormalizedOGD` learner, the scale trackers (winsorized EMA, two-timescale envelope, block median), the configs, and the scripts that reproduce every figure and number (including Table 1 and the GARCH surrogate). Data: the public Jane Street Real-Time Market Data Forecasting competition, plus MNIST-CNN gradient norms (Section 4).

A. Additional experimental tables and figures

Collected here for space: the leakage controls (Table 3), the single-window tuned comparison (Table 4), and the scale-drift/clipping mechanism figure (Figure 3). The ten-window replication (Table 2) and its reading remain in the body, since the across-window numbers—not the single-window tuning—carry the accuracy claims.

Table 3. **Leakage controls for the prequential** R^2 (date$[0, 120)$, 150k rows, causal standardization). The headline $R^2 \approx 0.28$ survives a shuffled-label null and is not a look-ahead artifact: it comes from within-day adaptation on a nonstationary stream, not from seeing the target or a transferable static fit.

Control	R^2	Reads as
Shuffled label (perm. y)	-0.00	no label leakage
Offline ridge (static / per-day oracle)	$0.02/0.18$	online 0.28: adaptation
Frozen w_T, forward window	-1.7	tracking, not transfer

Table 4. Best weighted R^2 (each method at its own tuned rate) on the continuous Jane Street stream, *causal* standardization, both round granularities, with the largest stable learning rate; each R^2 carries $\pm$one circular block-bootstrap standard error (block $\approx$ one trading day, 4000 resamples). Scale-*dependent* methods (top) are fragile (stable only to lr $\approx 10^{-2}$); the *bounded scale-free* family (bottom)—SN-OMD, normalized-GD, a fixed-τ clip (τ=20), and AdaGrad-Norm, all satisfying Theorem 3.1—stays bounded far higher and is far more accurate, with *none* dominating. These are single-window tuned numbers: the per-protocol leaders here (fixed-τ clip per-step, AdaGrad-Norm per-row) do *not* replicate across windows, where the robust leaders are normalized-GD (per-step) and SN-OMD with a block-median scale (per-row)—see Table 2. Interpretation, and the block-median tracker variant, are in Sections 4 and 4.2. R^2 is not comparable to competition scores (different protocol).

Method	R^2 per-row	R^2 per-step	Stable lr
Scale-dependent (fragile):			
OGD	$0.018_{\pm.006}$	$0.139_{\pm.012}$	$\leq 10^{-2}$
AdaptiveClip / RobustOMD	~ 0.01	—	$\leq 10^{-2}$
Scale-invariant (bounded; stable lr per row):			
Normalized-GD ($M \to 0$)	$0.197_{\pm.024}$	$0.316_{\pm.021}$	≤ 2
Scale-adaptive OGD ($M \to \infty$)	$0.162_{\pm.022}$	$0.080_{\pm.013}$	≤ 1
Fixed-τ clip (const M)	$0.246_{\pm.046}$	$\mathbf{0.382}_{\pm.047}$	≤ 0.2
AdaGrad-Norm	$\mathbf{0.329}_{\pm.030}$	$0.219_{\pm.014}$	≤ 10
SN-OMD (M=5, ours)	$0.285_{\pm.079}$	$0.309_{\pm.056}$	≤ 2

B. A per-regime budget heuristic

The achievable rate degrades smoothly with the tail. The core primitive is robust mean estimation: under a finite p-th moment the best estimator of a mean from n samples has error $\Theta(\sigma\, n^{-(1-1/p)})$ (Devroye et al., 2016; Lugosi & Mendelson, 2019), which interpolates the sub-Gaussian rate ($p = 2$) and no concentration ($p \to 1$). Aggregated online this yields, and is matched by, a regret of order $T^{1/p}$ (Zhang et al., 2026):

$$R_T \;\asymp\; \sigma \cdot T^{1/p}, \qquad T^{1/2}\;(p{=}2) \;\longrightarrow\; T\;(p \to 1). \tag{2}$$

Two refinements matter under nonstationarity. Partitioning $[T]$ into regimes r of length L_r, index p_r, and scale σ_r, and applying the bound within each, *static* regret (one comparator) is governed by the worst regime, while *dynamic* regret (a comparator per regime) has the per-regime costs *add*:

$$R_T^{\mathrm{stat}} \gtrsim \max_r \sigma_r L_r^{1/p_r}, \qquad R_T^{\mathrm{dyn}} \gtrsim \sum_r \sigma_r L_r^{1/p_r}. \tag{3}$$

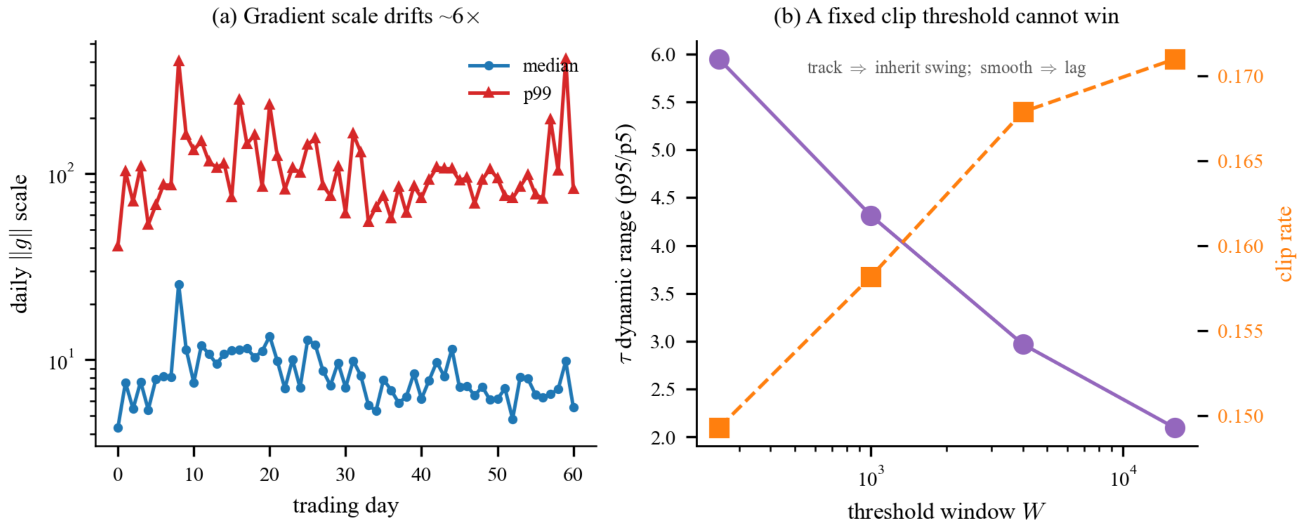

Figure 3. **Why scale-dependent clipping fails** (Section 4). (a) The gradient scale drifts $\sim 6\times$ across days. (b) A fixed-window adaptive clip threshold cannot win: a short window tracks the scale (so the step inherits the swing), a long window is smooth but lags regime onsets.

Because $L^{1/p}$ is convex, heavy-tail cost cannot be averaged across regimes: a few short heavy regimes ($p_r < 2$) dominate the sum. Three caveats keep (3) honest. First, it is a per-regime *budget*, not a proved regret lower bound at a fixed path length: it charges each regime its worst-case cost against a *fresh* comparator, so it lower-bounds dynamic regret only against an *unconstrained* comparator sequence—which is trivially $\Omega(T)$ in any case. The interesting object is regret at a given path length P_T, which (3) does not constrain. Second, it separates drift from the static case only qualitatively: prior heavy-tailed results fix a *single* p, a stationary scale, and a *static* comparator, and recover the $T^{1/p}$ rate (2) (Zhang et al., 2026; Zhang & Cutkosky, 2022; Bubeck et al., 2013), whereas here the per-regime costs *add* across a drifting exponent, and the static budget (a $\max$, not a sum) does not see this. Third—and this is why we keep (3) *qualitative*—the sum is dominated by regimes with $p_r < 2$, which we measure only on *pooled* gradients; after scale normalization the tail the learner actually sees has $p \approx 2$ (Section 4), and the super-$\sqrt{T}$ growth of the pooled budget largely evaporates. We therefore read (3) as a *motivating heuristic*—the interaction of tail drift with a moving comparator is what makes the dynamic problem hard—and not as a measured wall; a proper P_T-constrained Ω is left open. What SN-OMD delivers is to attain the per-regime worst-tail rate $T^{1/p}$ rather than diverge.

C. Dynamic-regret analysis

Beyond the unconditional stability of Theorem 3.1, SN-OMD admits a dynamic-regret bound under a condition on the scale tracker. We state it here (the proof is in Section D) and are explicit about its two soft spots—an assumption on the tracker, and a drift factor we measure to be a genuine rate on real data—which is why the body leads with the empirical results instead.

Assumption C.1 (Scale tracking)**.** The predictable scale brackets the local noise scale: $c_1\sigma_t \le s_t \le c_2\sigma_t$ for constants $0 < c_1 \le 1 \le c_2$. Taking $c_1 \le 1$ is without loss—it absorbs a constant factor into the cap M—and is the normalization Lemma D.2 uses.

Only the lower bracket is essential—it bounds $\|v_t\|$ and hence the rate, and under-estimating the scale is the sole divergence risk—while over-estimation merely shrinks the step η/s_t into mild underfitting, never divergence. Assumption C.1 is thus *one-sided in what matters*, which is what makes it dischargeable rather than an oracle (Proposition D.5, and Section 4.2 empirically); the upper bracket $s_t \le c_2\sigma_t$ is used only to bound the Freedman range constant of Theorem C.2 (any predictable almost-sure upper bound on s_t would serve), never the exponent. Define the tracker's upward variation $W_s = s_1 + \sum_t (s_t - s_{t-1})_+$ and the scale-weighted path $P_T^s = \sum_t s_t\|u_t - u_{t-1}\|$.

Theorem C.2 (Dynamic regret)**.** *Under Assumptions 2.1 and C.1, SN-OMD with $M = \Theta(T^{1/p})$ (or a doubling schedule,* *requiring no knowledge of* p*) satisfies, for any fixed comparator sequence* $u_{1:T}$*, with probability* $\geq 1-\delta$*,*

$$\begin{aligned} R_T(u_{1:T}) &= O\Big(\big(D\sqrt{W_s}+\sqrt{D\,P_T^s}\,\big)\sqrt{S_1}\,T^{(2-p)/(2p)}\sqrt{\log(1/\delta)} \\ &\qquad + (\sup_t \sigma_t)\,D\,T^{1/p}\log(1/\delta)\Big) \\ &= \tilde{O}\Big((\sup_t \sigma_t)\,(D+\sqrt{DP_T})\,T^{1/p}\log(1/\delta)\Big), \end{aligned} \tag{4}$$

where $S_1=\sum_t \sigma_t$*. The* $\sqrt{\log(1/\delta)}$ *multiplier and the additive* $\sup_t \sigma_t$ *term are the Freedman variance and range contributions (Lemma D.3; the curvature concentration of step (A) in Section D is of the same order), so the first line is a genuine high-probability bound. The second line is its* bounded-*scale simplification (*$W_s=O(1)$*, with* $S_1\leq T\sup_t\sigma_t$*);* ***this assumption fails on our data****, where* $W_s\sim T$ *(Section 4.2), so the* $T^{1/p}$ *form is per-regime—see the remark below.*

Remark C.3 (The drift is a genuine rate). The tail index p sets the exponent (through $M^{2-p}=T^{(2-p)/p}$, giving $T^{1/p}$), while the scale drift enters through $\sqrt{W_s}$. On a persistently fluctuating scale W_s accumulates *linearly* in T ($W_s\sim T$, verified in Section 4.2), so $\sqrt{W_s}\sim\sqrt{T}$ is a genuine rate: (4) is $T^{1/p}$ *per regime*, or $T^{1/p+1/2}$ over a horizon with a growing number of regimes—which at $p=2$ is the *vacuous* T, so the theorem's content is strictly per-regime. The tracker's update timescale controls the *constant* in W_s (Remark C.4), not the exponent. Escaping the $\sqrt{T}$ would need a switching bound in the regime count, left open.

Remark C.4 (Tracker design). A peak-hold envelope $s_t=\max\{c\cdot\mathrm{median}(\text{recent }\|g_t\|),(1-\rho)s_{t-1}\}$ has $W_s\approx\sup_t s_t+\rho\sum_t s_t$; choosing $\rho\asymp 1/L$ (the regime length) minimizes the W_s constant, whereas too large a ρ churns. An unknown regime length is handled by a strongly-adaptive wrapper (Daniely et al., 2015; Cutkosky, 2020) using SN-OMD's interval-regret property (Theorem 3.1 gives the required bounded feedback).

Remark C.5 (Scope: comparator and step schedule). Two limitations are worth stating plainly. *(i) Comparator.* The high-probability guarantee is per *fixed* (predictable) comparator sequence $u_{1:T}$: the martingale term needs u_t to be $\mathcal{F}_{t-1}$-measurable (Lemma D.3), so the Freedman event is comparator-specific and Theorem C.2 does not hold uniformly over all comparators without a union/chaining bound. In particular the *per-round minimizer* $u_t=\arg\min_u \ell_t(u)$ is $\mathcal{F}_t$-measurable, not predictable, and is *not* covered; our dynamic comparator must be fixed in advance with a given path length. *(ii) Step schedule.* Theorem C.2 is proven for the tuned constant step $\eta_t\equiv\eta$ (known horizon T, or a doubling schedule), whereas Algorithm 1 and Theorem 3.1 deploy the anytime $\eta_t=\eta/\sqrt{t}$ for horizon-free stability. These are the same update under two schedules; carrying the fixed-η regret to the $1/\sqrt{t}$ schedule is the standard anytime/doubling reduction, which we do not reproduce here.

D. Proofs

Throughout, $v_t=g_t/s_t$, $\hat{g}_t=\mathrm{clip}(v_t,M)$ with $M\geq 1$, and $p=\inf_t p_t$. (The regret bound takes $M=\Theta(T^{1/p})\geq 1$, the genuine-truncation regime; the $M\to 0$ endpoint of Section 4.1 is the classical normalized-GD, whose regret is standard and not re-derived here. Theorem 3.1, by contrast, holds for every $M>0$.) We use the filtration $(\mathcal{F}_t)$ with $\mathbb{E}[g_t\mid\mathcal{F}_{t-1}]=\bar{g}_t$ and s_t predictable ($\mathcal{F}_{t-1}$-measurable).

D.1. Stability (Theorem 3.1)

Since $\hat{g}_t=v_t\min\{1,M/\|v_t\|\}$, $\|\hat{g}_t\|\leq M$ for every realization. On a bounded domain $\Pi_{\mathcal{W}}$ keeps $\|w_t\|\leq D$. On $\mathbb{R}^d$, $\|w_{t+1}-w_t\|\leq(\eta/\sqrt{t})M$, so $\|w_t\|\leq\|w_1\|+M\eta\sum_{k<t}k^{-1/2}\leq\|w_1\|+2M\eta\sqrt{t}$. No moment assumption is used. □

D.2. Clipped moment and bias

Lemma D.1. *For* $v\in\mathbb{R}^d$*,* $M>0$*,* $p\in(1,2]$*:* $\|\mathrm{clip}(v,M)\|^2\leq M^{2-p}\|v\|^p$ *and* $\|v-\mathrm{clip}(v,M)\|\leq\|v\|^p/M^{p-1}$*.*

Proof. If $\|v\|\leq M$: $\mathrm{clip}(v,M)=v$ and $\|v\|^2=\|v\|^p\|v\|^{2-p}\leq\|v\|^pM^{2-p}$, while $v-\mathrm{clip}(v,M)=0$. If $\|v\|>M$: $\|\mathrm{clip}(v,M)\|^2=M^2=M^pM^{2-p}\leq\|v\|^pM^{2-p}$, and $\|v-\mathrm{clip}(v,M)\|=\|v\|-M\leq\|v\|=\|v\|^p\|v\|^{1-p}<\|v\|^pM^{1-p}$. □

Lemma D.2 (Moment and bias under Assumption C.1)**.** *There is a constant* κ *with* $\mathbb{E}[\|\hat{g}_t\|^2\mid\mathcal{F}_{t-1}]\leq M^{2-p}\kappa$*, and the bias* $b_t=\mathbb{E}[v_t-\hat{g}_t\mid\mathcal{F}_{t-1}]$ *obeys* $\|b_t\|\leq\kappa/M^{p-1}$*.*

Proof. Take conditional expectations in Lemma D.1 with $v = v_t$ and use $\mathbb{E}[\|g_t\|^{p_t} \mid \mathcal{F}_{t-1}] \le \kappa_0 \sigma_t^{p_t}$ and $s_t \ge c_1 \sigma_t$: $\mathbb{E}\|\hat{g}_t\|^2 \le M^{2-p}\mathbb{E}\|v_t\|^p \le M^{2-p}\kappa_0 c_1^{-p}$, and $\|b_t\| \le \mathbb{E}\|v_t\|^{p_t}/M^{p_t-1} \le \kappa_0 c_1^{-p_t}/M^{p-1}$ (using $M \ge 1$, $p_t \ge p$). Set $\kappa = \kappa_0 c_1^{-2}$, which dominates both $\kappa_0 c_1^{-p}$ and $\kappa_0 c_1^{-p_t}$ since $c_1 \le 1$ and $p, p_t \le 2$ (Assumption C.1). □

Because $M \ge 1$ and $p_t \ge p$, all uses of $M^{2-p_t}, M^{-(p_t-1)}$ above are bounded by the worst-case $M^{2-p}, M^{-(p-1)}$; the single-p bounds are thus valid for drifting p_t.

D.3. High probability

Lemma D.3 (Freedman step)**.** *Let (X_t) be a martingale difference sequence with $|X_t| \le b$ and $\sum_{t\le T}\mathbb{E}[X_t^2 \mid \mathcal{F}_{t-1}] \le V$ almost surely. Then w.p. $\ge 1-\delta$, $\sum_{t\le T} X_t \le \sqrt{2V\log(1/\delta)} + \frac{2b}{3}\log(1/\delta)$. In particular, for feedback Z_t with $\|Z_t\| \le B$ and $X_t = \langle Z_t - \mathbb{E}[Z_t \mid \mathcal{F}_{t-1}], w_t - u_t\rangle$ (u_t* predictable, *i.e. $\mathcal{F}_{t-1}$-measurable, so X_t is a martingale difference; $\|w_t - u_t\| \le D$), one may take $b = 2BD$ and $V = D^2\sum_t \mathbb{E}[\|Z_t\|^2 \mid \mathcal{F}_{t-1}]$.*

Proof. (X_t) is a martingale difference sequence ($\mathbb{E}[X_t \mid \mathcal{F}_{t-1}] = 0$). Freedman's inequality gives $\Pr(\sum_t X_t \ge \lambda) \le \exp(-\lambda^2/(2V + 2b\lambda/3))$; inverting the right-hand side to δ and using $\sqrt{x+y} \le \sqrt{x} + \sqrt{y}$ yields the bound. For the instantiation, $|X_t| \le \|Z_t - \mathbb{E}[Z_t \mid \mathcal{F}_{t-1}]\|\,\|w_t - u_t\| \le 2BD$ and $\mathbb{E}[X_t^2 \mid \mathcal{F}_{t-1}] \le D^2\mathbb{E}[\|Z_t\|^2 \mid \mathcal{F}_{t-1}]$ since $\mathbb{E}\|Z - \mathbb{E}Z\|^2 \le \mathbb{E}\|Z\|^2$. □

D.4. Varying-step telescope

Lemma D.4. *SN-OMD is OGD with step $\eta_t = \eta/s_t$ on the clipped raw gradient. For any comparator sequence $u_{1:T}$, writing $W_s = s_1 + \sum_t (s_t - s_{t-1})_+$ and $P_T^s = \sum_t s_t\|u_t - u_{t-1}\|$,*

$$\sum_{t=1}^{T} s_t\big(\|w_t - u_t\|^2 - \|w_{t+1} - u_t\|^2\big) \le D^2 W_s + 2D\,P_T^s.$$

Proof. Write $s_0 := 0$, so that $W_s = \sum_{t=1}^T (s_t - s_{t-1})_+$. Shifting the index in the subtracted term, $\sum_{t=1}^T s_t\|w_{t+1} - u_t\|^2 = \sum_{t=2}^{T+1} s_{t-1}\|w_t - u_{t-1}\|^2$, hence

$$\sum_{t=1}^{T} s_t\big(\|w_t - u_t\|^2 - \|w_{t+1} - u_t\|^2\big) = \sum_{t=1}^{T}\big(s_t\|w_t - u_t\|^2 - s_{t-1}\|w_t - u_{t-1}\|^2\big) - s_T\|w_{T+1} - u_T\|^2,$$

and the final boundary term is nonpositive and dropped. Splitting each summand into a comparator-move and a scale-change part,

$$s_t\|w_t - u_t\|^2 - s_{t-1}\|w_t - u_{t-1}\|^2 = s_t\big(\|w_t - u_t\|^2 - \|w_t - u_{t-1}\|^2\big) + (s_t - s_{t-1})\|w_t - u_{t-1}\|^2,$$

which is at most $2Ds_t\|u_t - u_{t-1}\| + (s_t - s_{t-1})_+ D^2$ by the reverse triangle inequality and $\|\cdot\| \le D$. Summing over t gives $2DP_T^s + D^2 W_s$. □

D.5. Dynamic regret (Theorem C.2)

Roadmap and where each assumption enters. The linearized regret splits into three sums—(A) optimization, (B) martingale, (C) clipping bias—each closed by the lemmas below. (A) uses the varying-step telescope (Lemma D.4) for the comparator/ scale-drift terms and the clipped second moment (Lemmas D.1 and D.2) for the curvature (whose random sum is concentrated by a second application of Lemma D.3); (B) uses the Freedman step (Lemma D.3); (C) uses the clipping-bias bound (Lemma D.2). The two assumptions enter in a few isolated places: Assumption 2.1 (finite p-th moment) only through Lemma D.2, which converts $\mathbb{E}\|g_t\|^{p_t} \le \sigma_t^{p_t}$ into the conditional second-moment and bias bounds of (A)–(C); Assumption C.1's *lower* bracket $s_t \ge c_1\sigma_t$ in the curvature of (A)/(B) and the bias of (C); and its *upper* bracket $s_t \le c_2\sigma_t$ in exactly one further place—the range constant b_A of the curvature concentration in (A) below (any predictable a.s. upper bound on s_t would serve), never the exponent. So over-estimating the scale is harmless to the *rate*, costing only that Freedman range constant. The dependency is *Lemma D.1* $\to$ *Lemma D.2* $\to \{$(A),(B),(C)$\}$, with Lemma D.4 feeding (A) and Lemma D.3 feeding (B); the final display balances (A)'s curvature against (C)'s bias through the choice of η and M.

Write $d_t := \mathrm{clip}(g_t, Ms_t) = s_t\hat{g}_t$ for the clipped *raw* gradient, so that SN-OMD is exactly OGD with adaptive step $\eta_t = \eta/s_t$ on d_t (Lemma D.4). By convexity $R_T(u_{1:T}) \le \sum_t \langle \bar{g}_t, w_t - u_t\rangle$, and we split the true subgradient as $\bar{g}_t = m'_t + \beta_t$, where $m'_t = \mathbb{E}[d_t \mid \mathcal{F}_{t-1}]$ and $\beta_t = \bar{g}_t - m'_t = \mathbb{E}[(g_t - d_t) \mid \mathcal{F}_{t-1}]$ is the (raw) clipping bias. Then

$$\sum_t \langle \bar{g}_t, w_t - u_t\rangle = \underbrace{\sum_t \langle d_t, w_t - u_t\rangle}_{\text{(A) optimization}} + \underbrace{\sum_t \langle m'_t - d_t, w_t - u_t\rangle}_{\text{(B) martingale}} + \underbrace{\sum_t \langle \beta_t, w_t - u_t\rangle}_{\text{(C) bias}}.$$

(A) The OGD one-step inequality with step η_t, summed and combined with Lemma D.4, gives $\sum_t \langle d_t, w_t - u_t\rangle \le \frac{1}{2\eta}(D^2 W_s + 2DP_T^s) + \frac{\eta}{2}\sum_t \|d_t\|^2/s_t$. Since $\|d_t\|^2 = \|\mathrm{clip}(g_t, Ms_t)\|^2 \le (Ms_t)^{2-p}\|g_t\|^p$ (Lemma D.1) and $s_t \ge c_1\sigma_t$ (Assumption C.1), the curvature sum has conditional mean $\le M^{2-p}\kappa' \sum_t \sigma_t = M^{2-p}\kappa' S_1$, with $S_1 = \sum_t \sigma_t$ and $\kappa' := \kappa_0 c_1^{1-p}$ an absolute constant (the moment/lower-bracket constants of Assumption 2.1 and Lemma D.2). The curvature sum $\sum_t \|d_t\|^2/s_t$ is itself random, so its conditional-mean bound must be promoted to hold with high probability, not used pathwise: apply Lemma D.3 a second time to the predictable-variance sequence $\xi_t := \|d_t\|^2/s_t - \mathbb{E}[\|d_t\|^2/s_t \mid \mathcal{F}_{t-1}]$. The clip gives $0 \le \|d_t\|^2/s_t \le M^2 s_t \le M^2 c_2 \sup_t \sigma_t =: b_A$ (Assumption C.1), and $\sum_t \mathbb{E}[\xi_t^2 \mid \mathcal{F}_{t-1}] \le b_A\, M^{2-p}\kappa' S_1 =: V_A$, so with probability $\ge 1-\delta$

$$\sum_t \|d_t\|^2/s_t \;\le\; M^{2-p}\kappa' S_1 + \sqrt{2V_A \log(1/\delta)} + \tfrac{2}{3} b_A \log(1/\delta).$$

Under $M = \Theta(T^{1/p})$ and the tuned η below, both deviation terms are of the same $T^{1/p}$ order as the mean after the η-balance and are absorbed into the $\sqrt{\log(1/\delta)}$ factor and additive range term of (4). Their *constant*, though, is worth displaying: relative to the conditional mean $M^{2-p}\kappa' S_1$, and writing $\bar{\sigma} := S_1/T$ for the mean scale,

$$\frac{\sqrt{2V_A \log(1/\delta)}}{M^{2-p}\kappa' S_1} = \Theta\Big(\sqrt{\tfrac{\sup_t \sigma_t}{\bar{\sigma}}\ \log\tfrac{1}{\delta}}\Big), \qquad \frac{\frac{2}{3} b_A \log(1/\delta)}{M^{2-p}\kappa' S_1} = \Theta\Big(\tfrac{\sup_t \sigma_t}{\bar{\sigma}}\ \log\tfrac{1}{\delta}\Big),$$

both $O(1)$ in the horizon—no change to the $T^{1/p}$ exponent—but with constant the *peak-to-mean scale ratio* $\sup_t \sigma_t/\bar{\sigma}$, entering *linearly* on the range branch. On a spiking scale this ratio is far from $O(1)$; it is large precisely because of the volatility clustering this paper studies (Section 4), so the price of a high-probability curvature bound is itself a form of the scale drift—a genuine constant cost, not a benign one. This is the step that makes term (A)—like (B)—a high-probability, not an in-expectation, bound. *(B)* The increments $\langle m'_t - d_t, w_t - u_t\rangle$ form a martingale difference with $\|d_t\| \le Ms_t$ and $\mathbb{E}[\|d_t\|^2 \mid \mathcal{F}_{t-1}] \le M^{2-p}\kappa'\sigma_t^2$; Lemma D.3 bounds this sum by $\tilde{O}(DM^{1-p/2}\sqrt{\sup_t \sigma_t \cdot S_1})$, the same order as (A). This step requires u_t predictable (Lemma D.3), so the Freedman event is comparator-specific; Theorem C.2 is accordingly a guarantee *per fixed* $u_{1:T}$, not one holding uniformly over all comparators (which would need a union or chaining bound over comparator sequences, not pursued here). *(C)* $\|\beta_t\| \le \mathbb{E}[\|g_t\|^p \mid \mathcal{F}_{t-1}]/(Ms_t)^{p-1} \le \kappa'\sigma_t/M^{p-1}$ (Lemma D.2), so (C) $\le D\kappa' S_1/M^{p-1}$. Choosing $\eta = \Theta\big(\sqrt{(D^2W_s + DP_T^s)/(M^{2-p}S_1)}\big)$ and $M = \Theta(T^{1/p})$ balances the curvature term of (A) against the bias (C) and yields

$$R_T(u_{1:T}) = O\Big(\sqrt{(D^2 W_s + DP_T^s)\, M^{2-p} S_1} + DS_1/M^{p-1}\Big),$$

which with $M = \Theta(T^{1/p})$ equals $(D\sqrt{W_s} + \sqrt{DP_T^s})\sqrt{S_1}\, T^{(2-p)/(2p)}$ up to the bias term. *If* the scale variation were bounded, $W_s = O(1)$, and using $S_1 \le T\sup_t \sigma_t$, this collapses to $\tilde{O}((\sup_t \sigma_t)(D + \sqrt{DP_T})T^{1/p}\log(1/\delta))$—but W_s is a data quantity, and we *measure* $W_s \sim T$ (Section 4.2). So the $\sqrt{W_s}$ factor is a genuine $\sqrt{T}$ rate and this $T^{1/p}$ simplification holds only *per regime*; globally the bound is $T^{1/p+1/2}$ (at $p=2$, the vacuous T). The stationary case ($\sigma_t \equiv \sigma$, $p = 2$, no drift so $W_s = \sigma$) recovers $\sigma(D + \sqrt{DP_T})\sqrt{T}$, the classical dynamic-OGD rate. □

D.6. Reductions

Unknown p: $M = \Theta(T^{1/p})$ needs p; a doubling schedule over M, using that the bound is unimodal in $\log M$ with a flat optimum, removes this at a $\mathrm{polylog}(T)$ cost (Zhang et al., 2026). *Unknown $\|u\|$ / unbounded domain:* Theorem 3.1 gives bounded feedback $\|\hat{g}_t\| \le M$, exactly the input required by the parameter-free potential of Zhang & Cutkosky (2022); substituting our $\hat{g}_t$ yields the unbounded-domain analogue with D replaced by $\|u\|$. *Unknown regime length:* Lemma D.4 plus the interval version of Lemma D.3 give SN-OMD an interval-regret guarantee on every $[a, b]$; a strongly-adaptive wrapper (Daniely et al., 2015; Cutkosky, 2020) then recovers the dynamic bound without knowing the regime structure, at a $\mathrm{polylog}$ overhead. A per-regime transition cost $O(D\,N\,\mathrm{polylog}\,T^{1/p})$ is incurred while the scale tracker re-concentrates after each of the N regime changes; it is dominated when regimes are macroscopic.

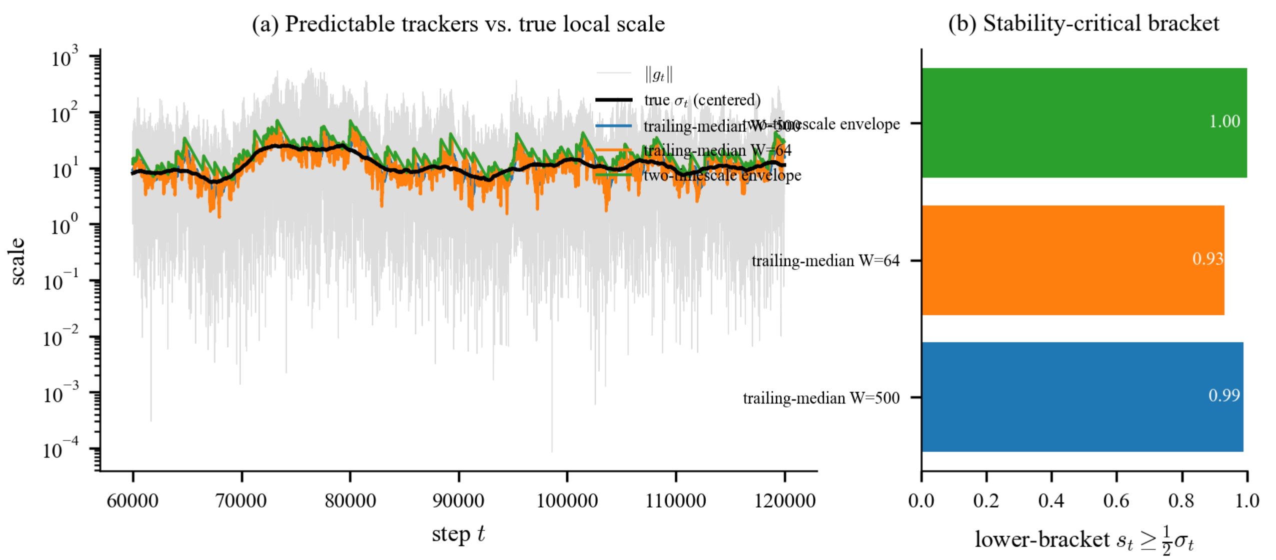

Figure 4. **Discharging Assumption C.1** (Section 4.2). A two-timescale "peak-hold" tracker on the real gradient-scale process (a) holds the lower bracket $s_t \geq \frac{1}{2}\sigma_t$ at every round while (b) attaining the smallest upward variation W_s/V_σ^+; a fast trailing median churns ($60\times$), a slow one lags and violates the bracket.

Table 5. Scale trackers on the real gradient-scale process (Section 4.2). The two-timescale envelope holds the lower bracket at 100% with the smallest upward variation W_s. V_σ^+ is the true upward scale drift.

Tracker	Lower bracket	W_s	W_s/V_σ^+
Trailing median, fast (W=64)	0.93	38.6k	$60\times$
Trailing median, slow (W=500)	0.99	5.0k	$7.9\times$
Two-timescale envelope (ours)	**1.00**	**4.2k**	**$6.6\times$**

D.7. Discharging Assumption C.1: a bracket guarantee

Assumption C.1 posits a predictable scale that brackets σ_t. We sketch why it is not an oracle: under a mild model of the drift (piecewise-slow variation plus within-regime mixing), the two-timescale envelope of Section 4.2 meets the essential lower bracket off a small exceptional set, so the burden moves from "an accurate scale estimate exists" to "the drift is piecewise slowly varying with a mixing scale," and the estimate is then *derived* from robust concentration. This is a sketch, not a theorem: the window-median step needs a blocking argument (below), which we state as an assumption rather than discharge.

D.8. Synthetic streams: known tail and the cap sweep

On synthetic streams with a controlled tail (Student-t index p; Figure 6), plain OGD's fitted cumulative-regret exponent rises $0.37 \to 0.70 \to 1.55$ as p falls $2 \to 1.3$—super-linear (divergent) once the tail is heavy—reproducing the real-data divergence with a *known* tail index, so the tail is the mechanism, not a learning-rate artifact. SN-OMD stays sublinear, with exponents *below* $1/p$ (the stochastic stationary instance is easier than the worst case, so this is consistent with, not a tightening of, Theorem C.2); against a drifting comparator its dynamic regret grows with path length at exponent ≈ 0.43, consistent with the $\sqrt{DP_T}$ term.

The cap sweep (Figure 7). Sweeping M at each cap's own tuned step turns the cap into a *mechanism test*: normalization already removes most of the tail ($\hat{\alpha}$: $2.4 \to 2.9$–3.7), so a finite cap should earn its keep only when the *normalized* gradient is still heavy. With an *imposed* heavy normalized tail ($p = 1.5$) the finite-M interior optimum beats the uncapped endpoint $\approx 240\times$ (heavy-tail instability) and the normalized-GD endpoint $\approx 10\times$ (magnitude-blind oscillation), both outside the seed-noise band; on Jane, where the normalized tail is indistinguishable from causal-estimation cost (Table 1), the same sweep has no interior optimum and the three cap-family members are comparable (Table 4). Both predictions of the

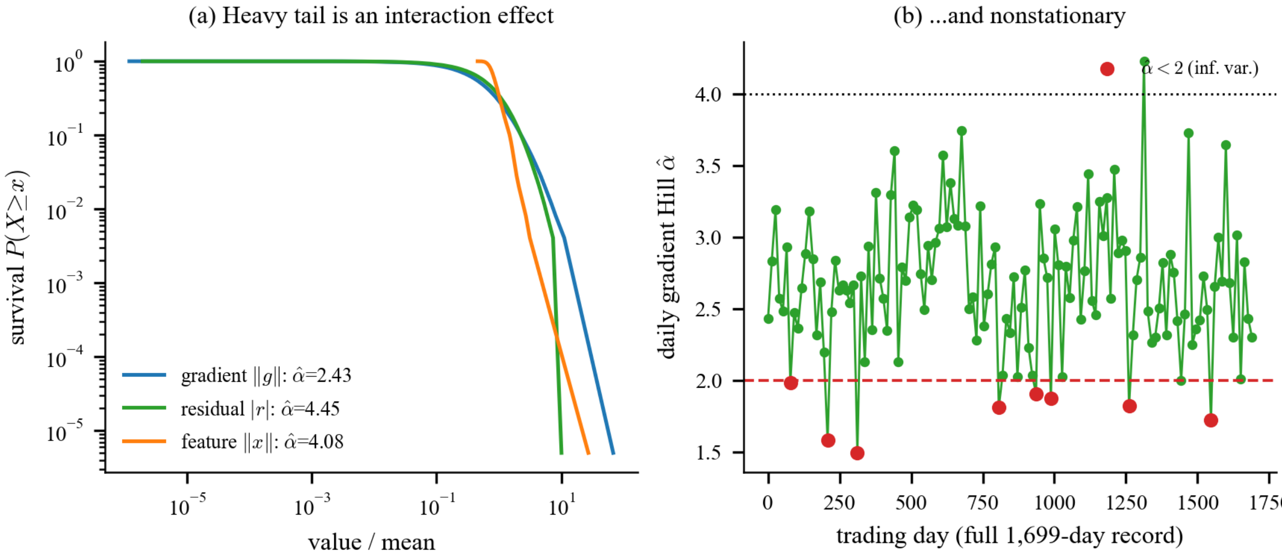

Figure 5. **The pooled tail is real, but conditionally removable** (Section 4). (a) The *pooled* gradient tail (Hill index $\hat{\alpha} \approx 2.43$) is heavier than either its residual or feature factor ($\hat{\alpha} \approx 4$)—an interaction effect from co-movement. (b) The daily gradient tail index across the full 1,699-day record drifts; per-day estimates occasionally fall below 2 but single-day Hill estimates are noisy. Dividing by a *predictable* causally-tracked scale lightens the tail to $\hat{\alpha} \approx 2.9$–3.7 (estimator-dependent; Table 1); a marginal-preserving order-shuffle leaves the pooled index unchanged (an order-statistic identity) yet abolishes the lightening, so what the tracker removes is serial dependence, not the marginal.

Table 6. The scale drift is a genuine rate, but its *constant* is controllable. Upward variation W_s vs. the tracker's update block length B (predictable robust median per block) on the 200,000-round gradient-scale process. Restricting the $W_s \sim T^{\beta}$ fit to horizons with $\gtrsim 50$ blocks gives $\beta \approx 1.0$–1.25 where reliably measurable (B=100, 10^3), with no monotone trend; the naive full-range fit's decline (to 0.17 at large B) is a finite-horizon artifact—B=10^4 has only $\sim$20 blocks and no $\geq$50-block horizon. What B controls is the constant, shrinking W_s by $\sim$5 orders of magnitude while roughly holding the lower bracket. Escaping the resulting $\sqrt{T}$ drift factor needs a switching bound in the regime count, left open.

block B	1	100	10^3	$3{\cdot}10^3$	10^4
W_s (full record)	1.8M	3216	319	108	29
β ($\geq$50 blk)	—	1.03	1.25	0.73	n/a
lower bracket	0.72	0.97	0.97	0.92	0.93

mechanism test are thus borne out. The cap therefore buys *accuracy* only when the tail survives normalization, and *stability unconditionally*: uncapped scale-adaptive OGD diverges at lr ≥ 2 while SN-OMD does not.

D.9. A second domain: image-model gradients

On the per-step gradient norm of a small CNN on MNIST (2,814 steps) the pooled tail is heavy ($\hat{\alpha} \approx 2.6$) and normalizing lightens it, but this is *schedule-driven*: within a constant-learning-rate window the tail is already light ($\hat{\alpha} \approx 6$–9), so the heaviness is scale decay, not within-phase drift. The within-regime effect—confirmed by the shuffle/iid-null controls and the feature–residual interaction—we establish only on the market data.

Drift model. Partition $[T]$ into N regimes by change points $t_0 < \cdots < t_N$. *Slow within-regime drift:* inside a regime the scale varies slowly, $\sigma_t/\sigma_{t-1} \in [1-\gamma, 1+\gamma]$ with $\gamma W \leq c_0$ for the window W. *Arbitrary jumps:* at the N change points σ_t may jump by any factor. This is the piecewise-slowly-varying picture the measured gradient scale fits ($\sim$30% within-day drift, occasional $3\times$ overnight jumps; Section 4).

Tracker. The envelope $s_t = \max\{c\,\widehat{m}_t,\ (1-\rho)s_{t-1}\}$, where $\widehat{m}_t = \mathrm{median}(\|g_{t-W}\|, \ldots, \|g_{t-1}\|)$ is the predictable short-window median and $\rho \asymp 1/L$ with $L = \min_r L_r$.

Proposition D.5 (Bracket guarantee, informal)**.** *Under Assumption 2.1, the drift model above, and β-mixing of the within-*

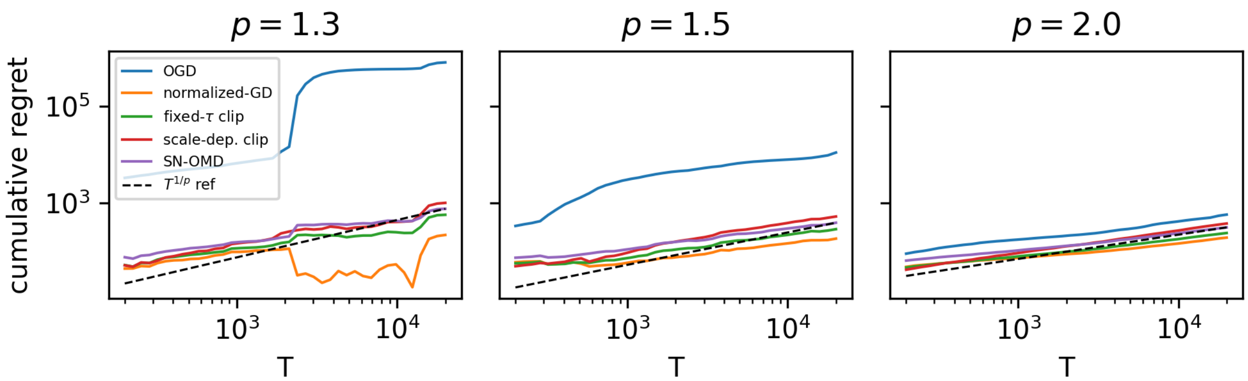

Figure 6. **Known-tail synthetic** (Section D.8). Cumulative regret vs. T at $p \in \{1.3, 1.5, 2.0\}$: OGD's exponent climbs past 1 (divergent) as the tail heavies, while SN-OMD stays sublinear and below the $T^{1/p}$ reference (dashed).

regime $\|g_t\|$ process (so a blocking argument yields window-median concentration—the iid order-statistic rate is otherwise only heuristic under the serial dependence documented in Section 4), for a large enough constant c, with probability $\geq 1-\delta$: (i, lower bracket) $s_t \geq c_1\sigma_t$ *for every t outside at most $O(W)$ rounds after each jump—an exceptional set of size $O(NW)$; and* (ii, bounded variation) $W_s = O(\sup_t \sigma_t + \rho \sum_t \sigma_t) = O(V_\sigma^+)$. *Consequently Theorem C.2 holds with one added term $O(D\,N\,W\,\sup_t \sigma_t)$, dominated whenever $NW \ll T^{1/p}$ (macroscopic regimes).*

Proof sketch (heuristic; the window-concentration step is the β-mixing assumption, not a discharged bound). Window concentration. On a window lying inside one regime, the population median of $\|g_t\|$ is a constant multiple of σ_t—a regular-variation/scale-family assumption on the conditional law, *beyond* the moment bound of Assumption 2.1. Under the within-regime β-mixing assumption a blocking argument gives empirical-median concentration around it—split the window into $\asymp W/\ell$ near-independent blocks of length $\ell \gtrsim$ the mixing time; the iid order-statistic bound is the ℓ=1 special case and is only heuristic under the serial dependence this paper documents. The slow drift shifts the constants by $O(\gamma W) = O(1)$. Hence $\widehat{m}_t \in [a\sigma_t, b\sigma_t]$ w.h.p., and $c \geq c_1/a$ gives $c\,\widehat{m}_t \geq c_1\sigma_t$, so the max meets the lower bracket whenever the window lies in one regime. *Jumps.* After an upward jump the window still holds pre-jump norms for $\leq W$ rounds; during that lapse the median term may under-estimate, giving the $O(NW)$ exceptional set. A downward jump only makes s_t conservatively high (the release lets it fall at rate ρ), never violating the lower bracket. *Variation.* Upward moves of s_t come only from $c\,\widehat{m}_t$ rising; between jumps these telescope to $\sup_t s_t$, and the geometric release contributes the churn $\rho \sum_t s_t$, so $W_s \leq \sup_t s_t + \rho \sum_t s_t$, which is $O(V_\sigma^+)$ for $\rho \asymp 1/L$ (Remark C.4). *Exceptional rounds.* On the $O(NW)$ lapse rounds the update is still bounded ($\|\hat{g}_t\| \leq M$, Theorem 3.1) but the clipping bias (C) is uncontrolled; charging it trivially at $O(D \sup_t \sigma_t)$ per round gives the added $O(D\,N\,W\,\sup_t \sigma_t)$.

The blocking argument is not free: each block must span several mixing times, so the window W needed for concentration inflates with the mixing coefficient, and with it the $O(NW)$ exceptional set. The proposition therefore says something only for regimes long enough that $NW \ll T^{1/p}$ *survives* this inflation; for short or rapidly-mixing regimes the lapse term is no longer dominated, and the guarantee degrades to the empirical discharge of Section 4.2. The residual—the post-jump lapse—is an additive term already of the form isolated in the Reductions above, and making ρ adaptive to an unknown L is exactly the strongly-adaptive wrapper of that subsection.

D.10. The deployed EMA vs. the analyzed envelope tracker

The regret analysis discharges Assumption C.1 with the two-timescale "peak-hold" envelope of Section 4.2, whereas the empirically strongest configuration uses a winsorized exponential moving average (EMA). Because a reviewer may reasonably ask whether the analyzed object and the deployed object are the same, we state precisely what does and does not depend on the choice of tracker.

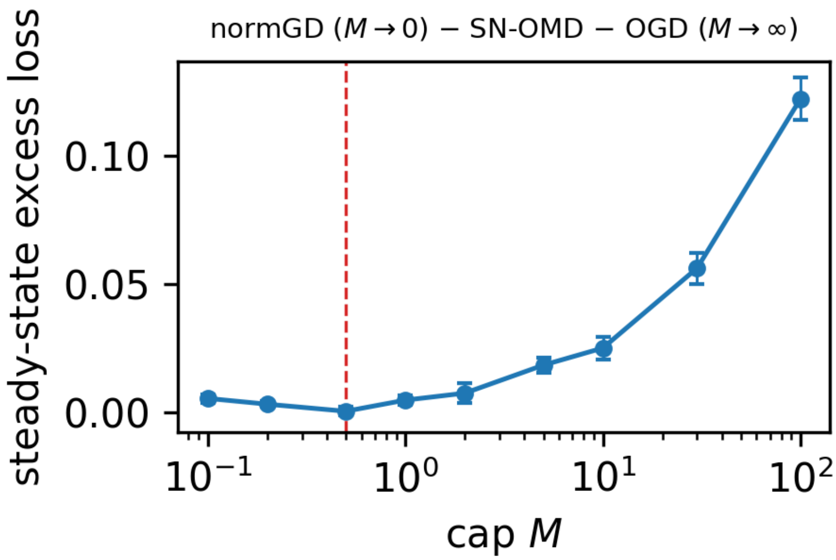

Figure 7. **The cap has an interior optimum** (synthetic, $p = 1.5$; Section 4.1). Steady-state excess loss vs. M, each cap at its own tuned step: the finite-M optimum beats both the normalized-GD end ($M \to 0$) and the uncapped scale-adaptive-OGD end ($M \to \infty$).

Stability is tracker-independent. Theorem 3.1 uses only $\|\hat{g}_t\| \le M$, which the cap enforces for *any* predictable $s_t > 0$. Non-divergence therefore holds verbatim for the EMA, for the envelope, and for any other tracker; the gap discussed here is never about stability, only about the leading regret constant.

Only the regret constant depends on the tracker. For any predictable s_t meeting the essential lower bracket $s_t \ge c_1\sigma_t$, the rate is $T^{1/p}$ (Theorem C.2); trackers differ only through the leading factor $\sqrt{W_s}$. The envelope is the tracker for which we can *prove* both the lower bracket and $W_s = O(V_\sigma^+)$ (Proposition D.5)—reacting up on a short robust median and releasing down only geometrically keeps every upward move a genuine rise (Table 5: 100% lower bracket, $W_s = 6.6\,V_\sigma^+$). A fast, *unwinsorized* EMA can in principle churn on heavy-tailed $\|g_t\|$ spikes (Table 5: $W_s = 60\,V_\sigma^+$); winsorizing the EMA at $k \cdot \mathrm{median}$ caps each upward step and is what keeps the deployed default's W_s finite in practice, though we do not give it a closed-form bound.

Which tracker to deploy: a three-way gap, widest at the best point. There are now three trackers in play and the accuracy/guarantee trade-off has three corners. *(1) The envelope* $s_t = \max\{c\,\hat{m}_t, (1-\rho)s_{t-1}\}$ is the only one with a *proven* bound: Proposition D.5 gives it both the lower bracket and $W_s = O(V_\sigma^+)$. *(2) The winsorized EMA* (deployed default) has no closed-form W_s bound—Proposition D.5 does not cover it, as its functional form is not the envelope's—only the empirical $W_s = 60\,V_\sigma^+$ of Table 5. *(3) The block median* ($B{=}10^4 \approx$ one day) is the most accurate *per-row* on our stream (per-timestep the constant-τ clip edges it, 0.382 vs. 0.351; Table 4) yet the least analyzed: Proposition D.5 does not cover it either, and we prove nothing about its W_s. So the best-performing tracker is the one with no guarantee; the gap between what we can *prove* (envelope) and what we would *deploy* (EMA, or block) is if anything wider than before. The ten-window replication (Table 2) makes this concrete: holding hyperparameters frozen from window 1, the block-median tracker cuts SN-OMD's per-row across-window R^2 standard deviation threefold ($0.11 \to 0.04$) and lifts its mean from 0.14 to 0.31—above every other bounded method and never negative—whereas the deployed winsorized-EMA is the variable one. This traces the $3\times$ block-bootstrap SE of Table 4 to the tracker rather than the cap, and sharpens the deploy question: the tracker that is most accurate and most regime-robust is exactly the one we cannot yet bound.

The accuracy gaps are easy to over-read, so we report block-bootstrap intervals on the aggregate gap, each method at its tuned rate (Section 4.2). The block tracker *does* beat the normalized-GD baseline per-row—0.38 vs. 0.197, gap +0.18, 95% CI $[+0.16, +0.21]$—but that edge is *protocol-specific*: under the batched per-timestep protocol it vanishes (0.351 vs. 0.316, gap +0.035, CI $[-0.05, +0.12]$). The mechanism is substitution, not coincidence: per-timestep batching averages the $\sim$9 correlated symbols in each $(\text{date}, \text{time})$ group, and that averaging is itself the variance reduction the smooth block scale supplies per-row, so the two do not compound—when the protocol smooths the update, a smoother tracker adds nothing. The same averaging is what lifts normalized-GD ($0.197 \to 0.316$, it had no such smoothing per-row) while, uncapped, it *enlarges* the untruncated step and halves scale-adaptive OGD ($0.162 \to 0.080$, Section 4); the block tracker's edge sitting exactly where the protocol lacks the smoothing is the same story from the tracker side. Against the EMA *default* its edge is *not* significant on either protocol (per-row gap $+0.10$, CI $[-0.04, +0.23]$). So we do *not* claim the block tracker is uniformly more accurate, and we keep the winsorized EMA as the reported default: it is the standard reactive choice, and the block tracker does not beat it significantly. Past the tuned range every tracker *degrades in loss* without diverging in the sense of Theorem 3.1—at $\mathrm{lr}{=}10$ the peak rolling loss reaches ≈ 30 (block) or ≈ 230 (EMA) while the $O(\sqrt{t})$ iterate bound still holds—so the earlier "pick either without risking divergence" was loose: the iterate never blows up, but usability does past tuning, the block tracker most gracefully.

This may look in tension with Section 4.2/Figure 2, where long windows are penalized for lagging regime onsets. The tension is only apparent, and it is exactly the distinction between the two uses of the scale. When the scale is a *divisor* ($\hat{g}_t = g_t/s_t$), a lagging s_t merely mis-normalizes the step magnitude—the divisor cancels first-order and a slightly stale scale costs little, so a slow block median is tolerable and even helpful (it is smoother and higher-bracket). We measured this rather than asserting it: sweeping the tracker's block timescale over $B \in \{50, \ldots, 20{,}000\}$ (a $400\times$ window range) moves the divisor's best-tuned weighted R^2 only within $[0.363, 0.386]$ (spread 0.022), if anything *rising* with the window—so a longer, laggier scale does not hurt the normalizer. When the scale is instead a *threshold* (clipping $\|g_t\|$ at s_t), the same sweep says something *sharper* than "long windows hurt thresholding." The threshold use never leaves $R^2 \approx 0.003$ at *any* window: clipping $\|g_t\|$ at a tracked scale is scale-*dependent* OGD, stable only at a tiny step ($\mathrm{lr} \approx 3 \times 10^{-4}$), so it is unusable no matter how the scale is tracked. Window choice is therefore moot on the threshold side—there is no usable operating point to tune toward—while the divisor is usable and window-insensitive across the whole $400\times$ range. This is the paper's central fragility finding (Section 4) arriving from the tracker-design direction: normalization decouples the step from the scale and thresholding does not, so the very lag that would wreck a threshold (it clips hardest exactly when the scale jumps up, discarding the informative post-onset gradients—the Section 4.2 clip-rate/dynamic-range tension) merely re-normalizes a divisor. The one-day block tracker is a normalizer, so no contradiction arises.

Closing the gap. Either run the envelope in deployment, accepting the small accuracy cost, or prove a W_s bound for the winsorized EMA under a heavy-tailed scale process: the winsorization already supplies the per-step cap the envelope gets from its median, so what remains is to control the geometric-decay churn term $\rho \sum_t s_t$, exactly as in Remark C.4, with ρ the EMA rate. The one residual modeling assumption—ρ matched to an unknown regime length—is removed by the strongly-adaptive wrapper of the preceding subsection.